

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [3]: df = pd.read_csv('Iris.csv')
# df = sns.load_dataset('iris')
```

```
In [4]: df.head()
```

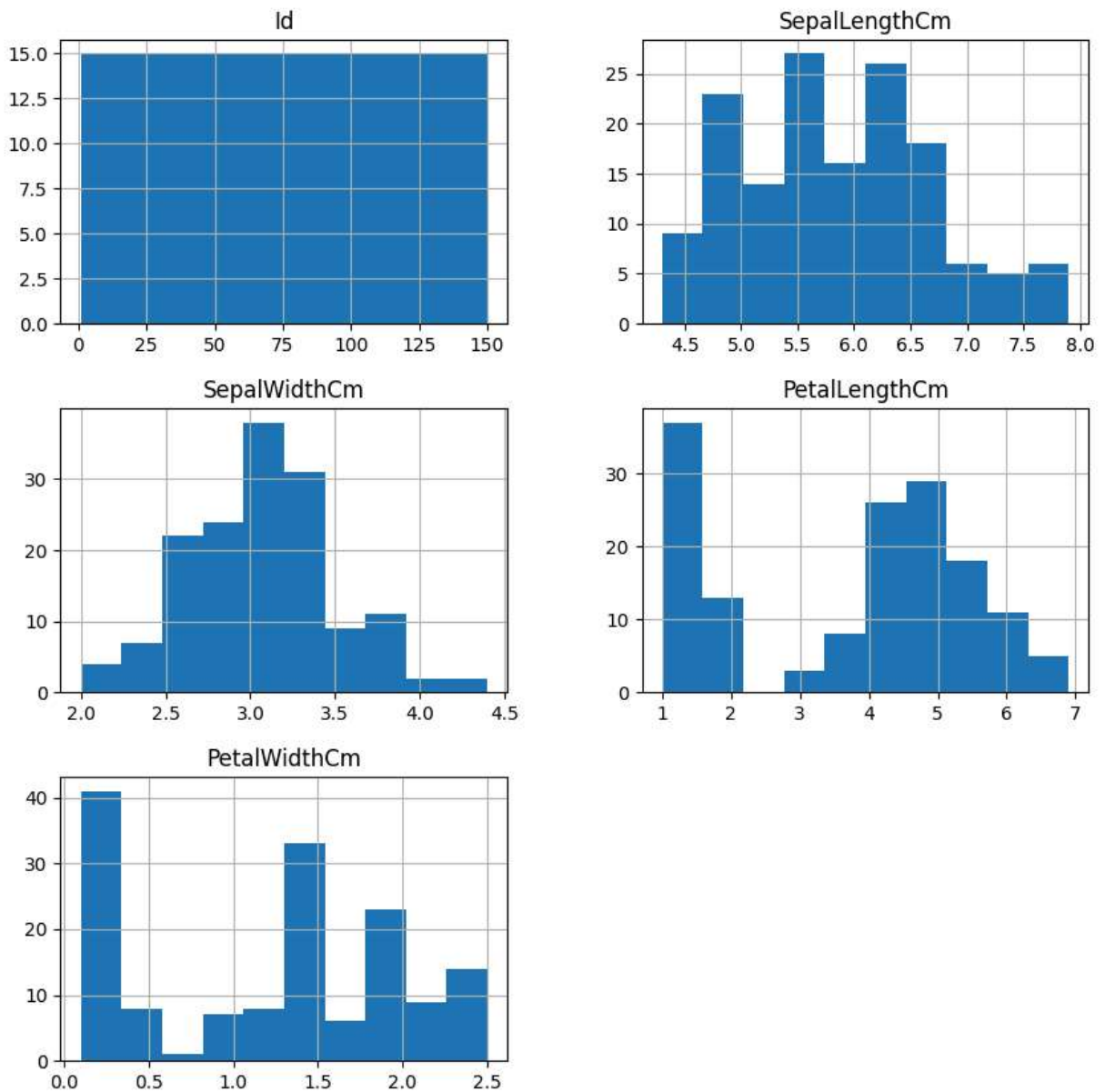
```
Out[4]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [5]: df.dtypes
```

```
Out[5]: Id                int64
SepalLengthCm          float64
SepalWidthCm           float64
PetalLengthCm          float64
PetalWidthCm           float64
Species                object
dtype: object
```

```
In [6]: df.hist(figsize=(10,10))
plt.show()
```



This figure contains:

Histograms of Iris dataset features

Each small graph represents distribution of one feature.

What Does X-axis Show?

In histogram:

X-axis shows feature values (data values)

Example:

- Sepal length values
 - Petal width values
-

What Does Y-axis Show?

Frequency (count)

Meaning:

- Number of observations falling in that range
-

Understanding Each Histogram

1. Id Histogram

Top-left graph.

X-axis

Shows:

Id values
from:

1 to 150

Y-axis

Shows:

how many records fall in each interval

Since IDs are evenly distributed:

- Histogram looks uniform.
-

2. SepalLengthCm Histogram

X-axis

Shows:

Sepal Length values
approximately:

4.3 cm to 7.9 cm

Y-axis

Shows:

number of flowers having those sepal
lengths

Observation

Most flowers have sepal length around:

5 to 6.5 cm

3. SepalWidthCm Histogram

X-axis

Shows:

Sepal Width values
approximately:

2 cm to 4.5 cm

Observation

Most values concentrated around:

2.8 to 3.5 cm

4. PetalLengthCm Histogram

X-axis

Shows:

Petal Length values
approximately:

1 cm to 7 cm

Important Observation

Histogram shows:

clustering

Meaning:

- Different flower species have different petal lengths

This helps classification.

5. PetalWidthCm Histogram

X-axis

Shows:

Petal Width values
approximately:

0.1 cm to 2.5 cm

Observation

Different groups visible.

Meaning:

petal width strongly separates species

Important Histogram Theory

What Are Bars Showing?

Each bar represents:

frequency of values in an interval

Example: If many flowers have sepal length between:

5.0-5.5

that bar becomes taller.

What Are Bins?

Histogram divides data into intervals called:

bins

Each bar = one bin.

Example

Suppose bin:

5.0-5.5

contains:

20 flowers

Then bar height becomes:

20

Most Important Viva Answer

Q. What does X-axis represent in histogram?

"X-axis represents feature values or intervals of the variable."

Q. What does Y-axis represent?

"Y-axis represents frequency or count of observations."

Overall Inference from Entire Figure

You can write:

"The histograms show distribution of different Iris dataset features. X-axis represents feature values and Y-axis represents frequency. Sepal features show approximately normal distribution, while petal features show clear clustering, indicating that petal measurements are more useful for species classification."

Very Important Observation

PetalLengthCm and PetalWidthCm histograms show:

multiple peaks

This indicates:

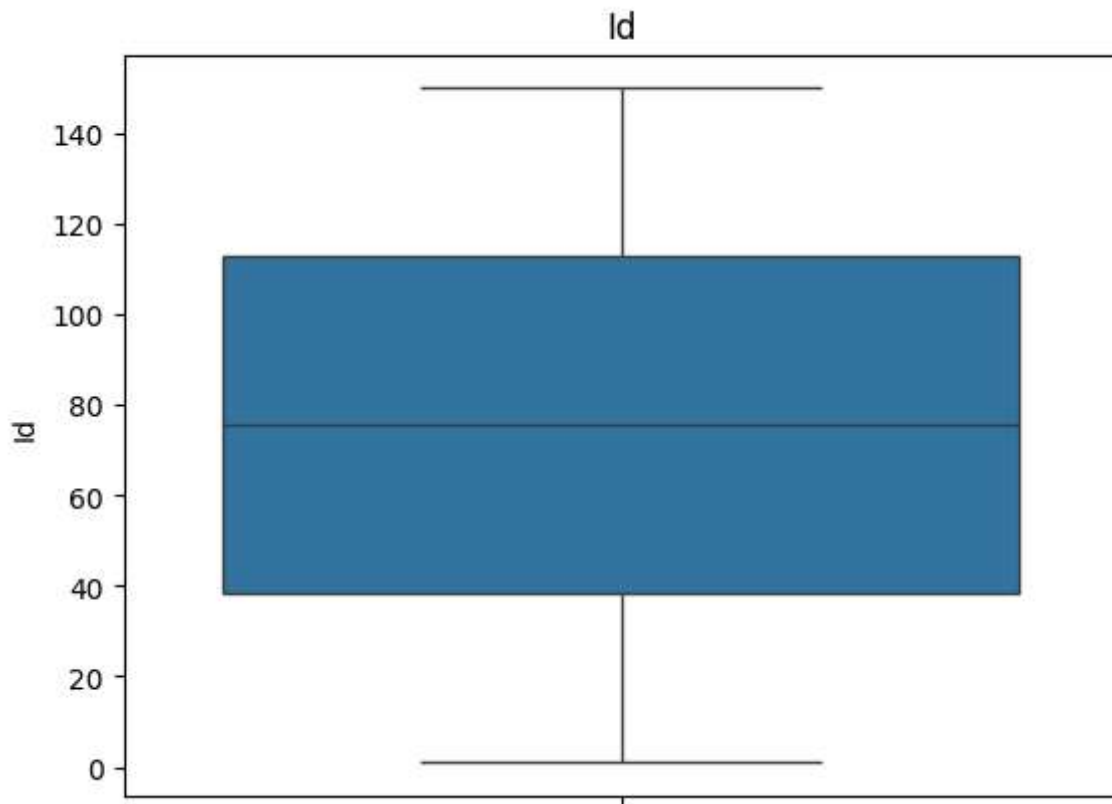
different species groups

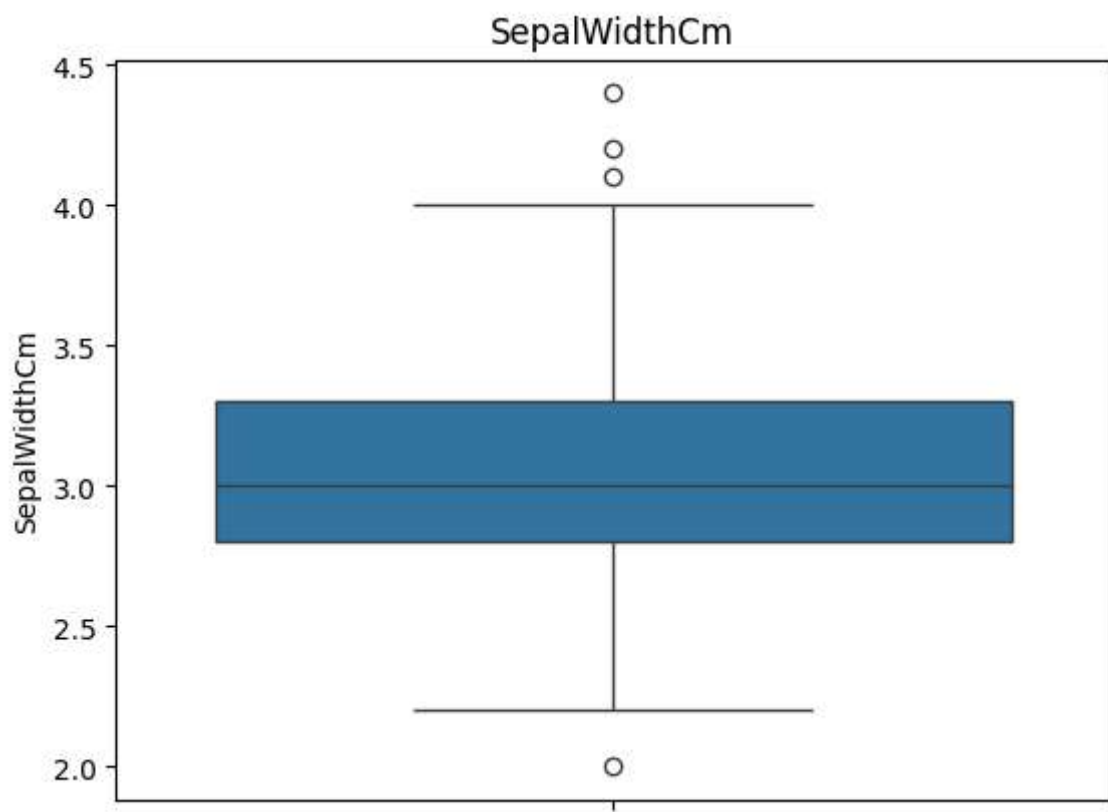
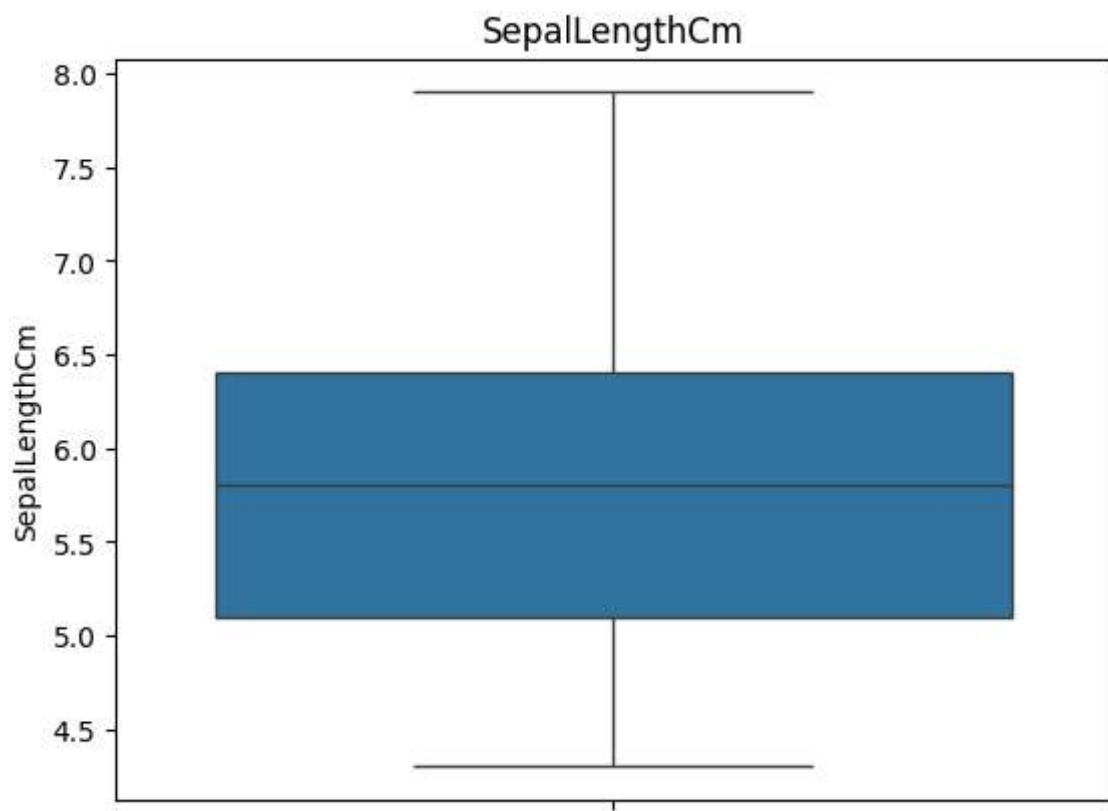
This is called:

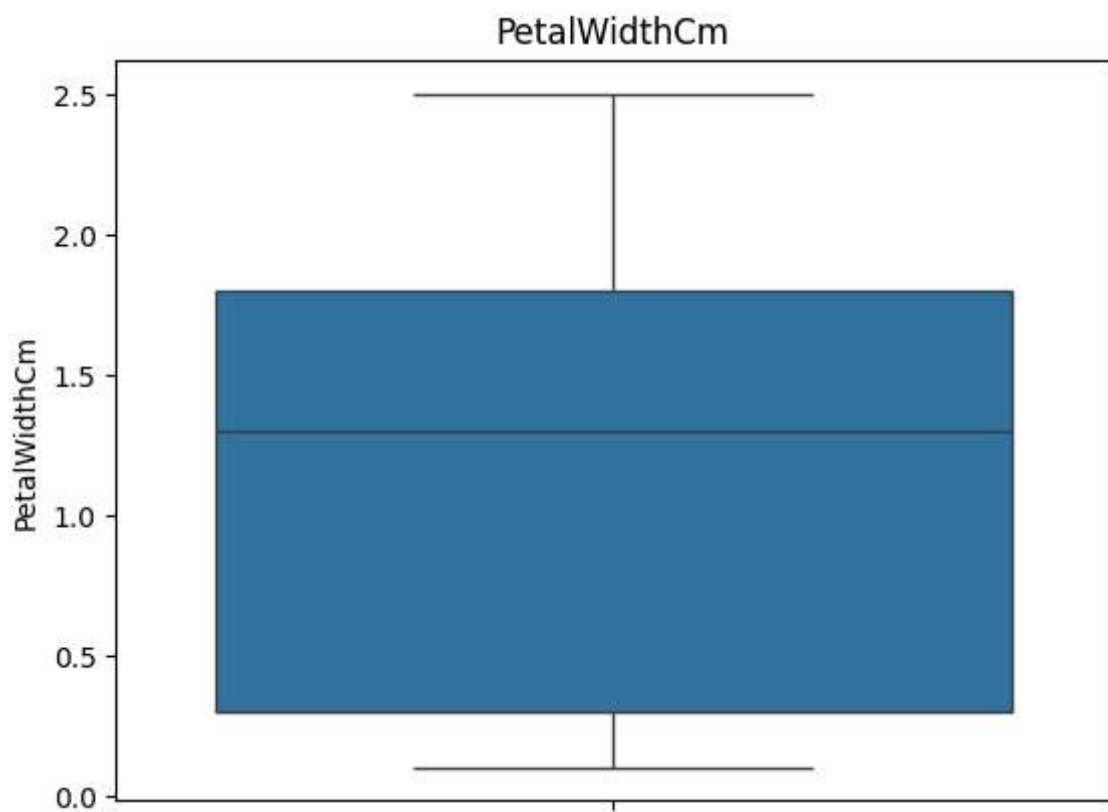
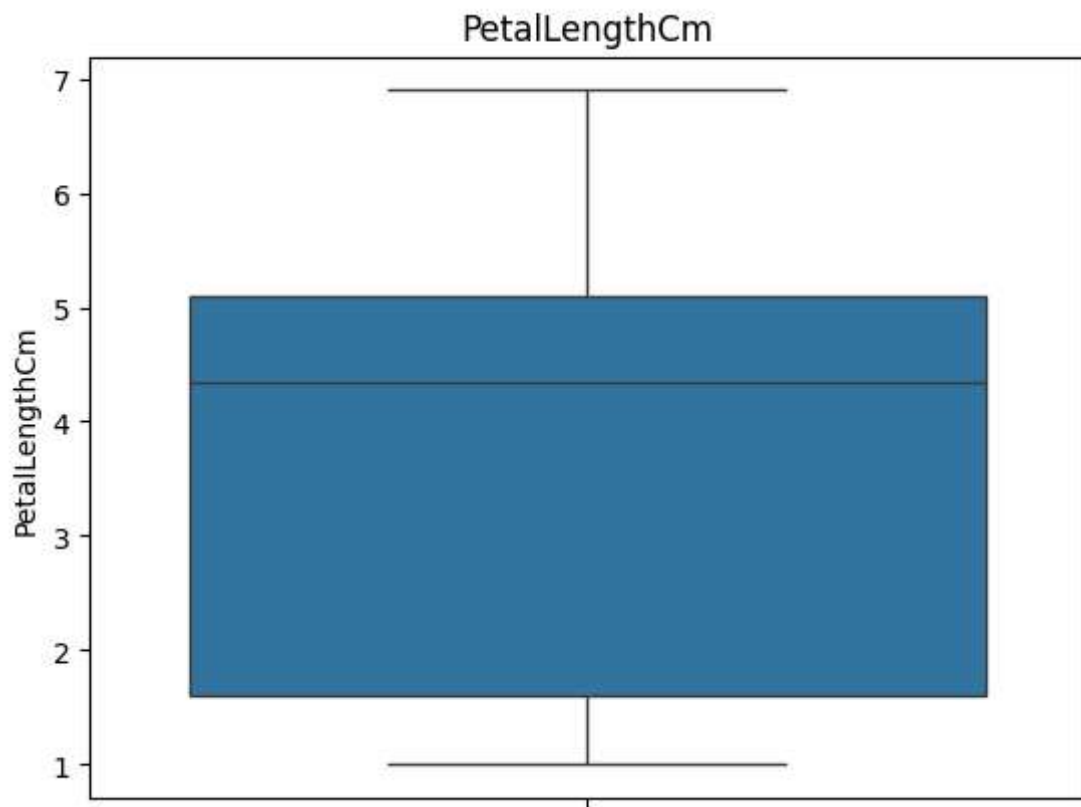
multimodal distribution

Very impressive viva point.

```
In [7]: for column in df.columns[:-1]:  
        sns.boxplot(y=df[column])  
        plt.title(column)  
        plt.show()
```







DSBDA Practical 10 — Data Visualization III

Iris Dataset Visualization using Histogram and Boxplot

Dataset Source: [UCI Iris Dataset](#)

Aim

To:

1. Analyze features and datatypes
 2. Plot histogram for each feature
 3. Plot boxplot for each feature
 4. Compare distributions and identify outliers
-

About Iris Dataset

The Iris dataset contains measurements of iris flowers.

Classes:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Features:

- Sepal Length
 - Sepal Width
 - Petal Length
 - Petal Width
-

Step 1: Import Libraries

Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Explanation

Library	Purpose
pandas	Data handling
seaborn	Visualization
matplotlib	Plot display

Step 2: Load Dataset

Code

```
df = sns.load_dataset('iris')
```

Display Dataset

Code

```
df.head()
```

Step 3: Check Dataset Information

Code

```
df.info()
```

Output Interpretation

You will observe:

- 150 rows
 - 5 columns
-

Features and Types

Feature	Type
sepal_length	Numeric
sepal_width	Numeric
petal_length	Numeric
petal_width	Numeric
species	Nominal/Categorical

Answer for Question 1

How many features are there?

There are:

5 features

Types of Features

Numeric Features

- sepal_length
- sepal_width
- petal_length
- petal_width

Nominal Feature

- species
-

What is Numeric Data?

Data represented using numbers.

Example:

5.1, 3.5

What is Nominal Data?

Categorical data without order.

Example:

setosa, versicolor

Step 4: Create Histograms

Code

```
df.hist(figsize=(10,10))  
  
plt.show()
```

Explanation

```
df.hist()
```

Creates histogram for all numeric features.

```
figsize=(10,10)
```

Controls graph size.

```
plt.show()
```

Displays plots.

What Histograms Show

Histograms display:

- Distribution

- Frequency
 - Spread
 - Skewness
-

Observations from Histograms

1. Sepal Length

- Approximately normal distribution
- Most values between:

5 to 7

2. Sepal Width

- Slightly skewed
- Most values around:

2.5 to 3.5

3. Petal Length

Shows clear clustering.

Reason:

- Different species have different petal sizes
-

4. Petal Width

Also shows clustering and separation.

Useful for:

classification

Important Inference

Petal features separate species better than sepal features.

Step 5: Create Boxplots

Code

```
for column in df.columns[:-1]:  
    sns.boxplot(y=df[column])  
    plt.title(column)  
    plt.show()
```

Explanation

```
df.columns[:-1]
```

Selects all numeric columns except species.

```
sns.boxplot()
```

Creates boxplot.

Why Loop Used?

To create:

separate boxplot for each feature

Understanding Boxplot

Components

Component	Meaning
Box	IQR
Middle line	Median
Whiskers	Normal range
Dots	Outliers

IQR Formula

$$\text{IQR} = Q3 - Q1$$

Outlier Limits

$$\text{Lower\ Limit} = Q1 - 1.5(\text{IQR})$$

$$\text{Upper\ Limit} = Q3 + 1.5(\text{IQR})$$

Observations from Boxplots

1. Sepal Width Contains Outliers

You may observe:

dots outside whiskers

Meaning:

- Outliers present
-

2. Petal Length and Width

Less outliers and better grouping.

3. Median Comparison

Petal measurements show larger variation between species.

4. Spread Comparison

Features with larger boxes:

higher variability

Important Inference

Petal features are more useful for distinguishing flower species.

Theory

What is Histogram?

Graph showing frequency distribution.

What is Boxplot?

Graph showing:

- median
- quartiles
- spread

- outliers
-

Difference Between Histogram and Boxplot

Histogram	Boxplot
Frequency distribution	Statistical summary
Shows shape	Shows spread/outliers

What are Outliers?

Values far from normal data range.

What is IQR?

Difference between:

Q3 and Q1

What is Median?

Middle value of sorted data.

Important Viva Questions

Basic Questions

How many features in Iris dataset?

5 features.

Which features are numeric?

Sepal and petal measurements.

Which feature is nominal?

Species.

What does histogram show?

Frequency distribution.

What does boxplot show?

Median, spread, and outliers.

Intermediate Questions

Which feature contains outliers?

Usually sepal_width.

Why petal features important?

Better species separation.

What are whiskers?

Normal data range.

What are dots outside whiskers?

Outliers.

Advanced Questions

Why use boxplot instead of histogram?

Better for identifying outliers.

Which feature best separates species?

Petal length and petal width.

What indicates higher variability?

Larger IQR / wider box.

External May Ask Modifications

Plot Histogram for One Feature

```
sns.histplot(df['sepal_length'])  
  
plt.show()
```

Plot Boxplot with Species

```
sns.boxplot(x='species', y='petal_length', data=df)  
  
plt.show()
```

Add KDE to Histogram

```
sns.histplot(df['petal_width'], kde=True)  
  
plt.show()
```

Pairplot

```
sns.pairplot(df, hue='species')  
  
plt.show()
```

Why Pairplot Useful?

Shows:

- relationships between features
 - species separation
-

Common Exam Errors

Error	Reason
Wrong column name	Case-sensitive
Species included in histogram	Non-numeric data
plt.show missing	Plot not displayed

2-Minute Viva Explanation

"In this practical, we analyzed the Iris dataset using visualization techniques. First, we loaded the dataset and identified feature types. The dataset contains four numeric features and one nominal feature called species. We created histograms to study feature distributions and observed that petal features show better separation among species. Then we created boxplots to analyze spread and detect outliers. Sepal width showed some outliers, while petal features had clearer grouping. Histograms and boxplots are useful for understanding data distribution, variability, and outliers."